PCEOT402
ELECTRICAL MACHINES

Course Objectives:

At the end of the course the student will be able to

Explain the working and analyse the performance of DC machines, transformers, synchronous machines and induction machines

Electrical Machines

- Module1 : DC Machine
- Module 2: Transformer
- Module 3: Synchronuous Machine
- Module 4: Induction Machine

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the constructional details of DC machines and analyse the performance DC generator under various load conditions	**K3**
CO2	Explain the working and analyse the performance DC motor under various load conditions	**K3**
CO3	Analyse the performance of 1-phase and 3-phase transformers and auto-transformers	**K3**
CO4	Analyse the performance of synchronous generator under various load conditions	**K3**
CO5	Explain the working and starting methods of synchronous motor	**K2**
CO6	Describe the constructional details and analyse the performance of 3-phase and 1-phase induction motors.	**K3**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Constructional details of dc machines - armature winding - lap and wave (concepts only) (1 hour) DC Generator - principle of operation – emf equation – numerical problems (1 hour) Classification of DC generator – separately excited & self-excited – steady-state equations – numerical problems (1 hour) DC shunt generator - no-load (open circuit) characteristics – critical field resistance, critical speed, voltage buildup - load characteristics – armature reaction - cross magnetising & demagnetising effect (concepts only) (3 hours)	12
	Power flow diagram – losses and efficiency – maximum efficiency – numerical problems (1 hour) DC motor – back emf – torque equation – numerical problems (1 hour) Classification of DC motors – steady-state equations – numerical problems (1 hour) Characteristics of shunt and series motors (1 hour) Starting of DC motors (concepts only) Power flow diagram – losses and efficiency – numerical problems (1 hour) Testing - Swinburne's test – numerical problems (1 hour)	

2	Single phase transformers – constructional details - principle of operation - EMF equation - ideal and practical transformer – numerical problems (2 hours) Operation on no load and on load - phasor diagram at different load conditions - equivalent circuit - voltage regulation – numerical problems (2 hours) Losses and efficiency - maximum efficiency – numerical problems (2 hours) Testing of transformers - OC test, SC test – numerical problems (1 hour) Autotransformer – saving of copper – numerical problems (1hour) 3- phase transformer – construction - different connections of 3-phase transformers - Y-Y, Δ-Δ, Y-Δ, Δ-Y - numerical problems (2 hours)	**10**

3	Principle of Operation of 3-phase synchronous generator – classification - constructional features - EMF equation – coil-span factor and distribution factor (sinusoidal flux distribution only) – numerical problems (3 hours) Synchronous generator on no-load – open circuit characteristics – synchronous generator on load – armature reaction – effect of armature reaction (2 hours) Equivalent circuit - phasor diagram – voltage regulation – predetermination of voltage regulation by emf and mmfmethod – numerical problems (2 hours) Parallel operation - synchronous generator on infinite bus-bar – conditions – methods of synchronisation – dark lamp method - bright	**11**

	lamp method (2 hours) Synchronous motor – rotating magnetic field (no derivation) - principle of operation – starting methods (2 hours)	
4	3-phase induction motor – principle of operation - classification - constructional features – torque equation - torque-slip characteristics – numerical problems (3 hours) Phasor diagram - equivalent circuit (1 hour) Power flow diagram - losses and efficiency – numerical problems (1 hour) No-load and blocked-rotor tests – determination of equivalent circuit parameters – numerical problems (1 hour) Starting of induction motors – types of starters – DOL starter, autotransformer starter, star-delta starter, rotor resistance starter (no design) – numerical problems (3 hours) Single-phase induction motors – double revolving field theory –torque slip characteristics – types – split-phase, capacitor-start induction-run, permanentcapacitor types – applications (2 hours)	**11**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

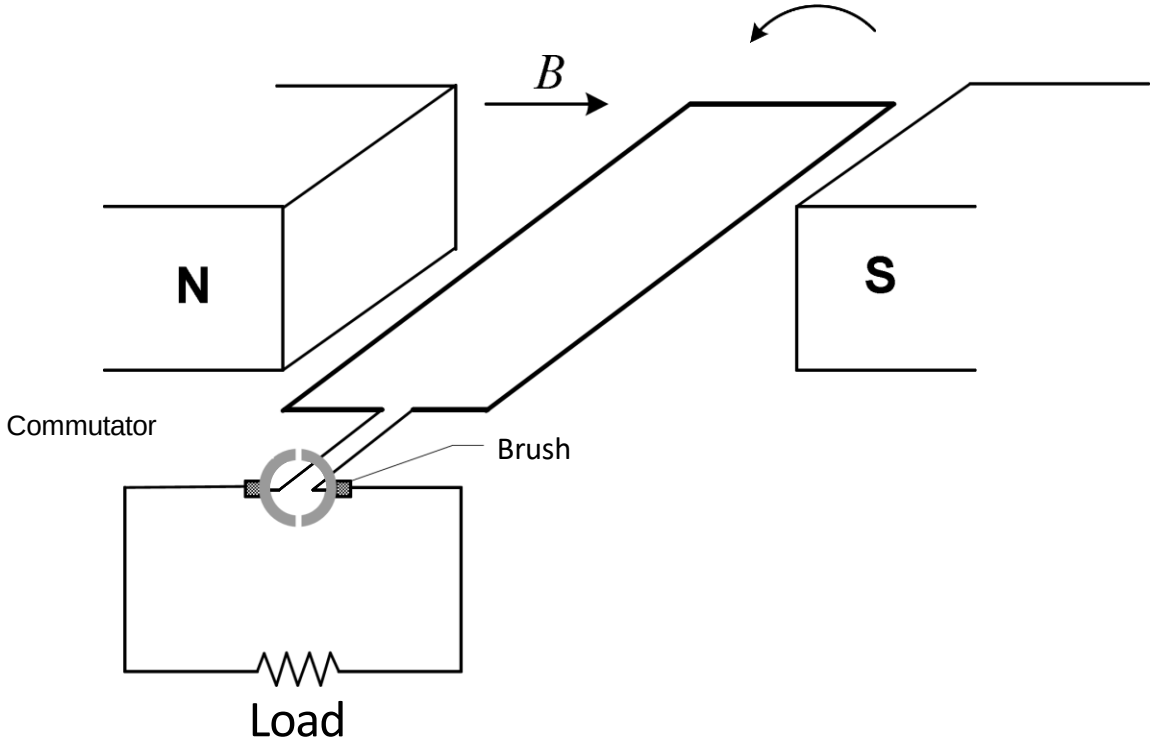
Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electrical Machinery	P.S. Bimbhra	Khanna Publishers	7th edition 2021
2	Electric Machines	D P Kothari & I J Nagrath	Tata McGraw Hill	5th edition 2017
3	Theory & Performance of Electrical Machines	J.B. Gupta	S K Kataria	15th edition 2022

Module 1
DC Machines

- Constructional details of dc machines - armature winding - lap and wave (concepts only) (1 hour)
- DC Generator - principle of operation – emf equation – numerical problems (1 hour)
- Classification of DC generator – separately excited & self-excited – steady-state equations – numerical problems (1 hour)

DC Generator - principle of operation

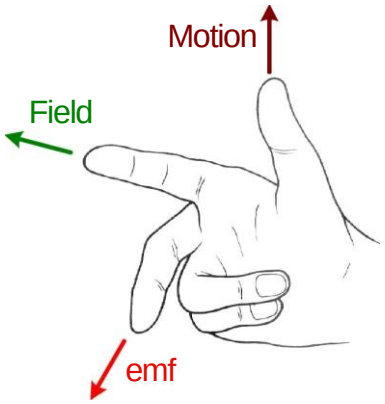
Basic Principle of DC Machines



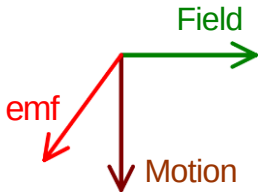
- Alternating voltage is created in the single turn coil when it rotates in the magnetic field
- The commutator rotates along with the coil and hence help to convert ac voltage into dc
- Brushes are stationary and collect the current form the commutator

Same machine can work as generator and motor

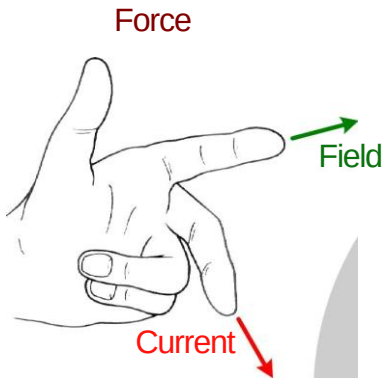
Generator



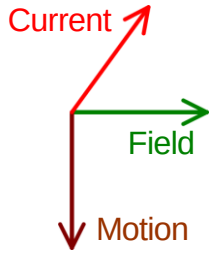




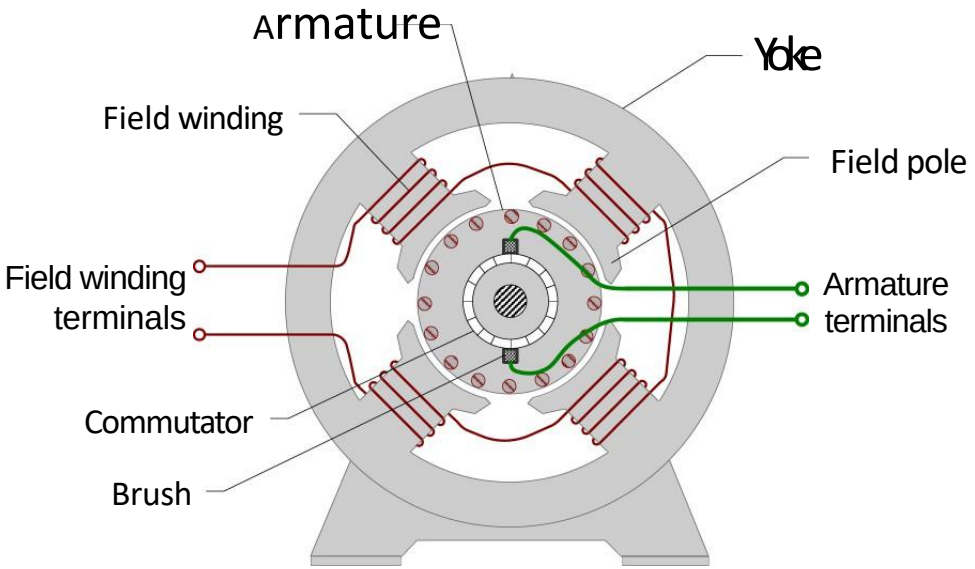
Motor





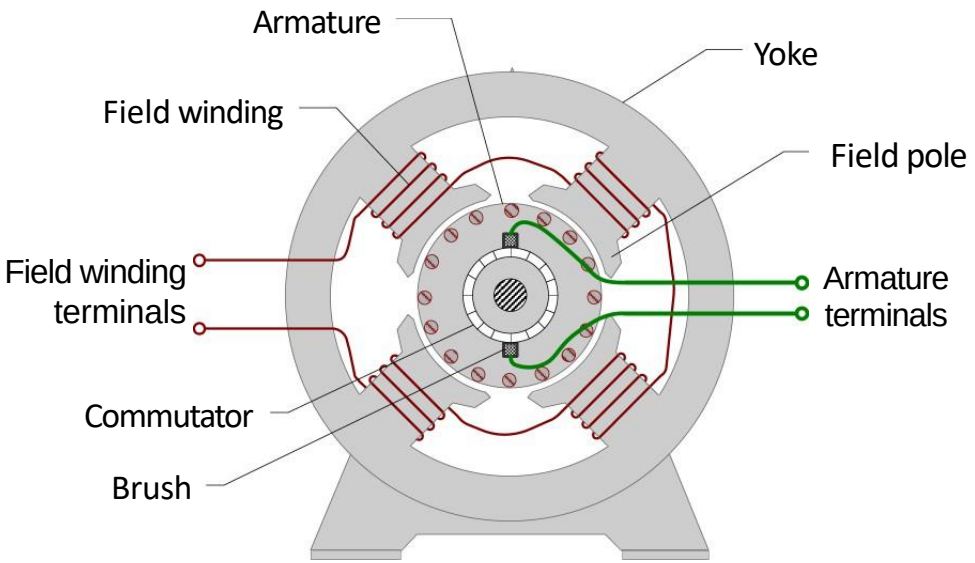


Basic Features of DC Machines



- Has a stator and a rotor
- Has a set of field poles and an armature
  With a distributed winding
- Field poles are normally on the stator side
- Filed poles may be of permanent magnet or electromagnet
- Require a commutator which converts ac voltage/current in the armature coils to dc in the external circuit

Parts of DC Machines



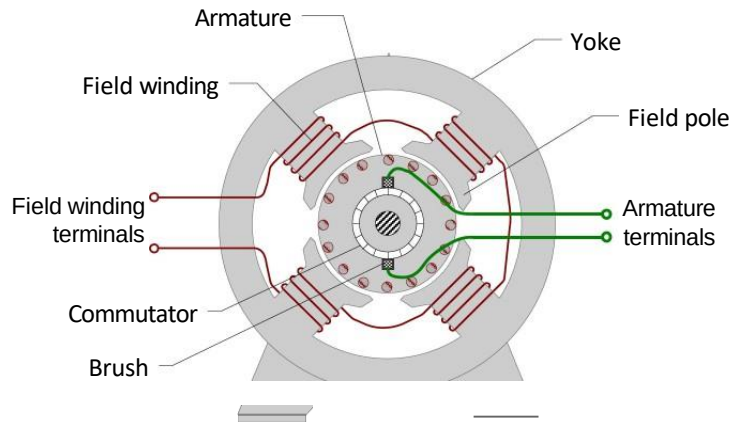
Yoke

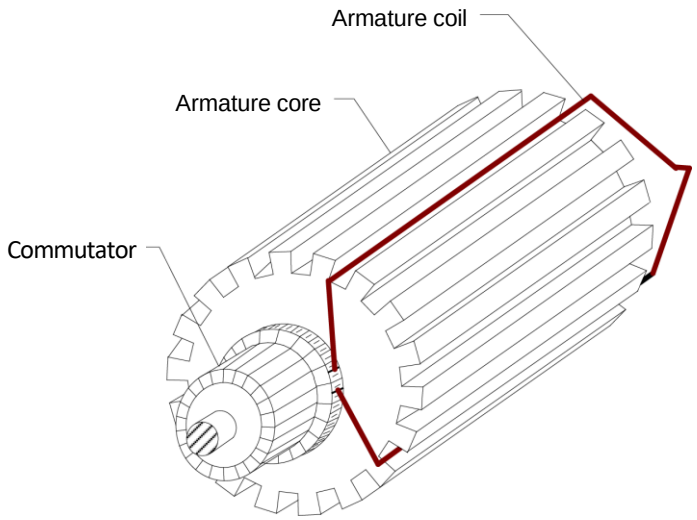
- Yoke is the outer frame of a dc machine
- It carries the magnetic flux produced by the field poles and completes the magnetic circuit
- It also serves the purpose of mechanical support to the poles and acts as a protective cover for the machine
- Yoke is made of steel with high permeability

Field Poles

- Field poles produce the required flux
- Comprises of pole core and pole shoe
- Field windings are placed over the pole core
- Pole shoe distributes the flux over the armature

Parts of DC Machines





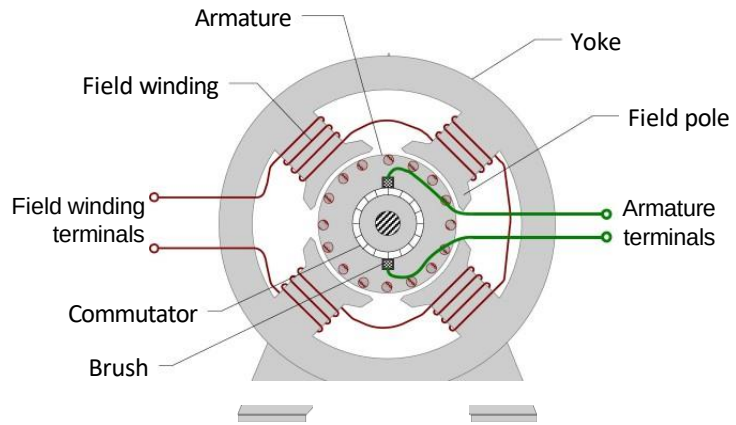
Armature Core

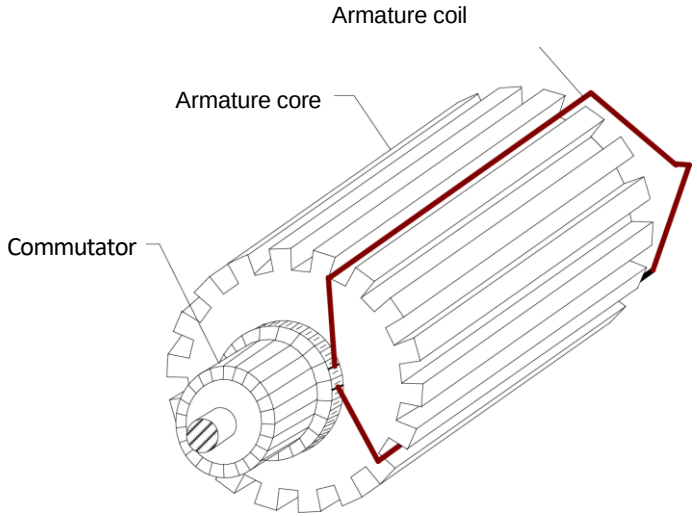
- Is a cylindrical core carrying the armature conductors
- Made of thin laminations
- The armature coils are placed in the slots

Commutator

- Commutator is a cylindrical structure attached to the armature
- Convert the alternating current induced in armature conductors into unidirectional current
- Made of copper segments which are insulated between each other by thin layers of mica

Parts of DC Machines





Brush

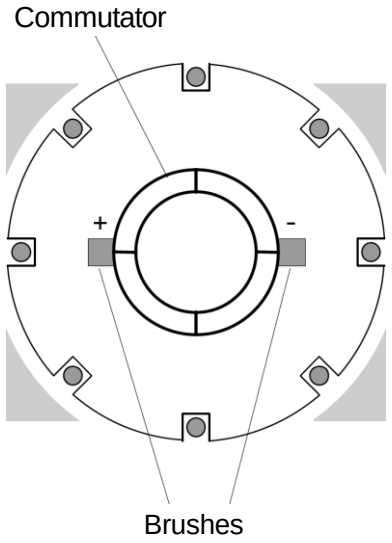
- Collect current from the rotating commutator
- Usually made of rectangular carbon blocks
- Held in brush holders attached to the machine's frame

Armature Winding

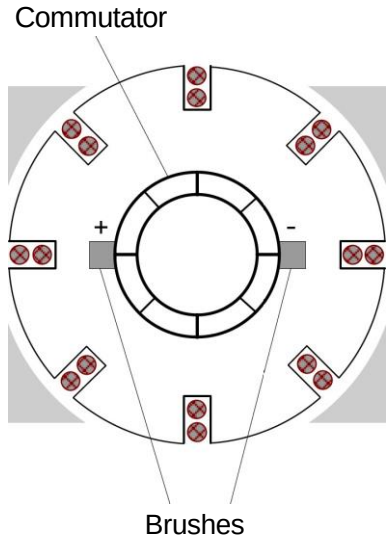
- Armature winding is a string of coils connected together
- The armature conductors are placed in the slots in a single layer or multiple layers.

Layers in winding

Single Layer Winding

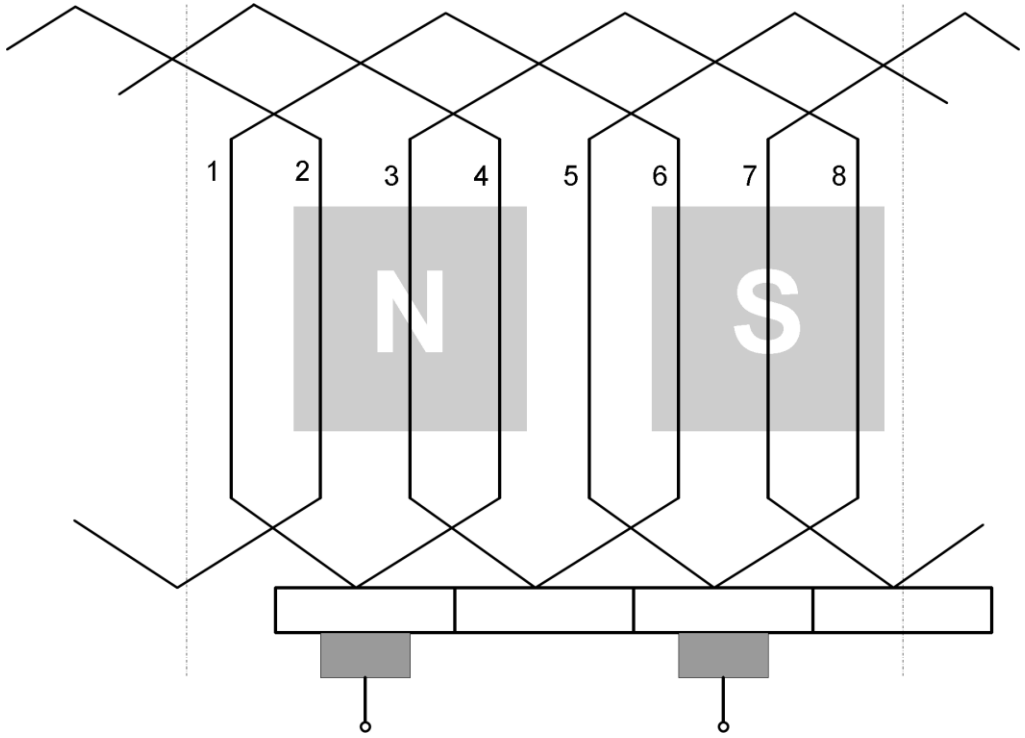


Two Layer Winding

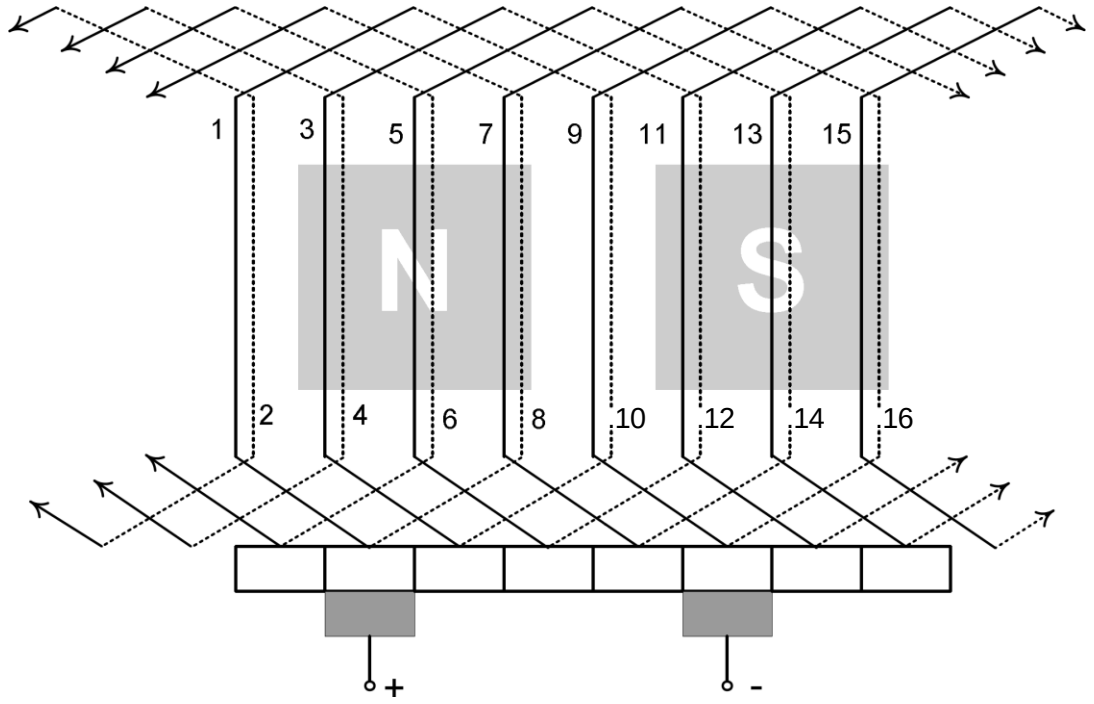


Developed Winding Diagram

Single Layer Winding

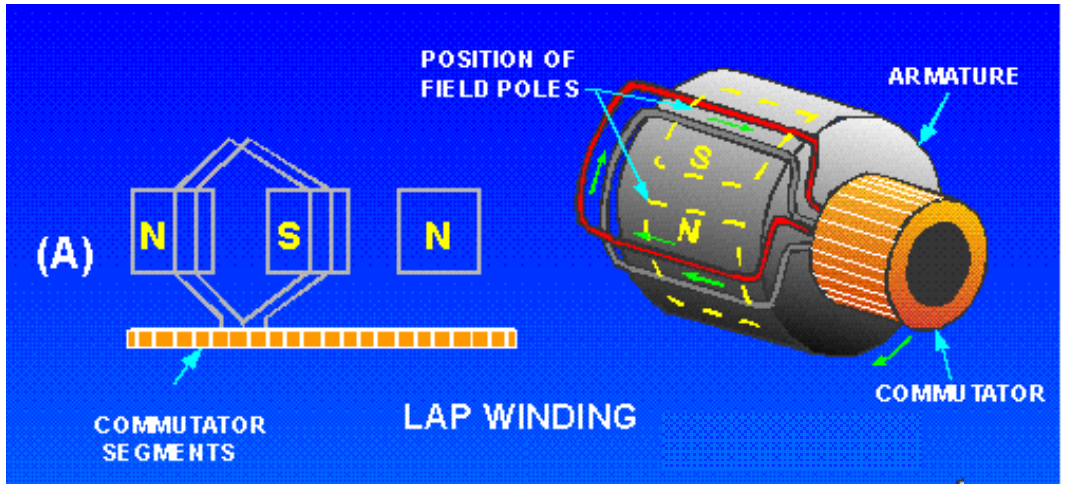


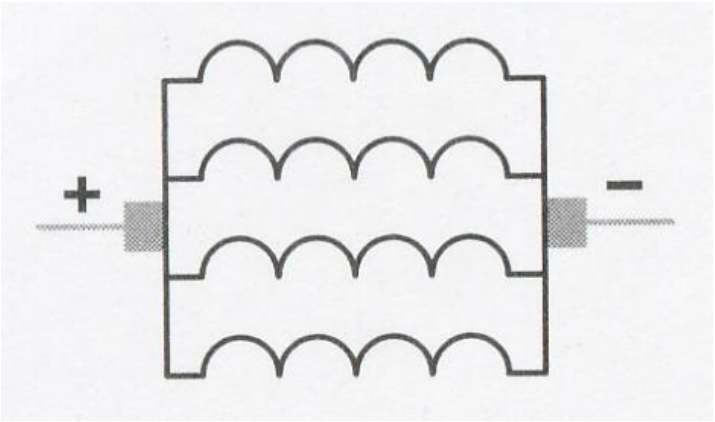
Two Layer Winding



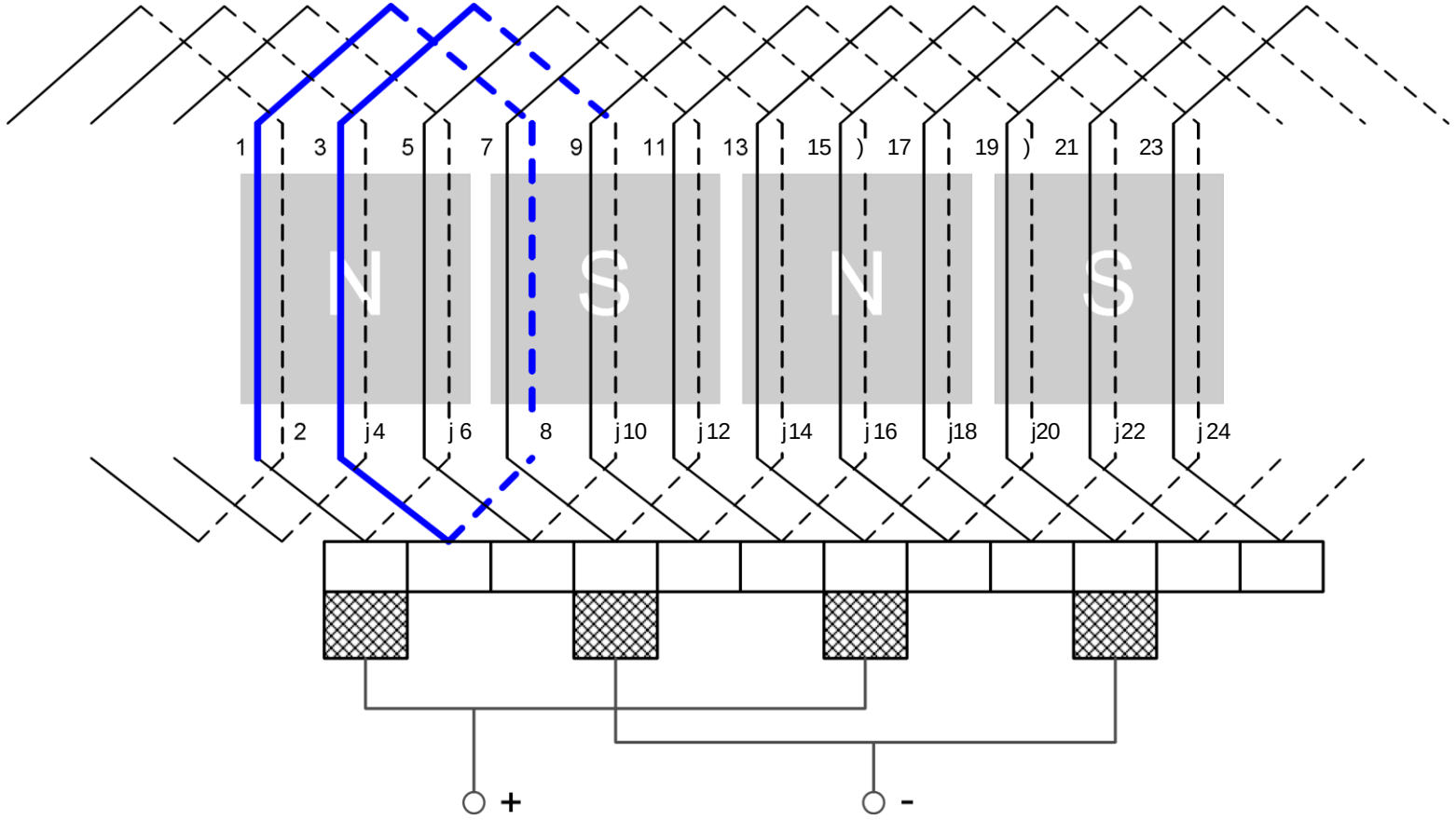
ARMATURE WINDINGS

- Lap Wound Armatures
  - are used in machines designed for low voltage and high current
  - armatures are constructed with large wire because of high current
  - Eg: - are used is in the starter motor of almost all automobiles
  - The windings of a lap wound armature are connected in parallel. This permits the current capacity of each winding to be added and provides a higher operating current
  - No of current path, **C=2p** ; p=no of pair of poles

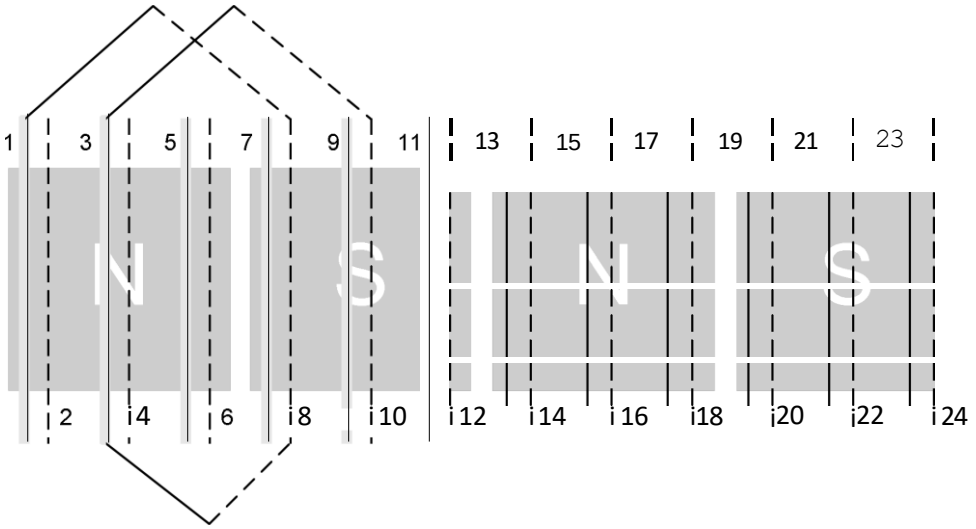




Lap Winding



Lap Winding Example



4 pole 12 slot double layer lap winding

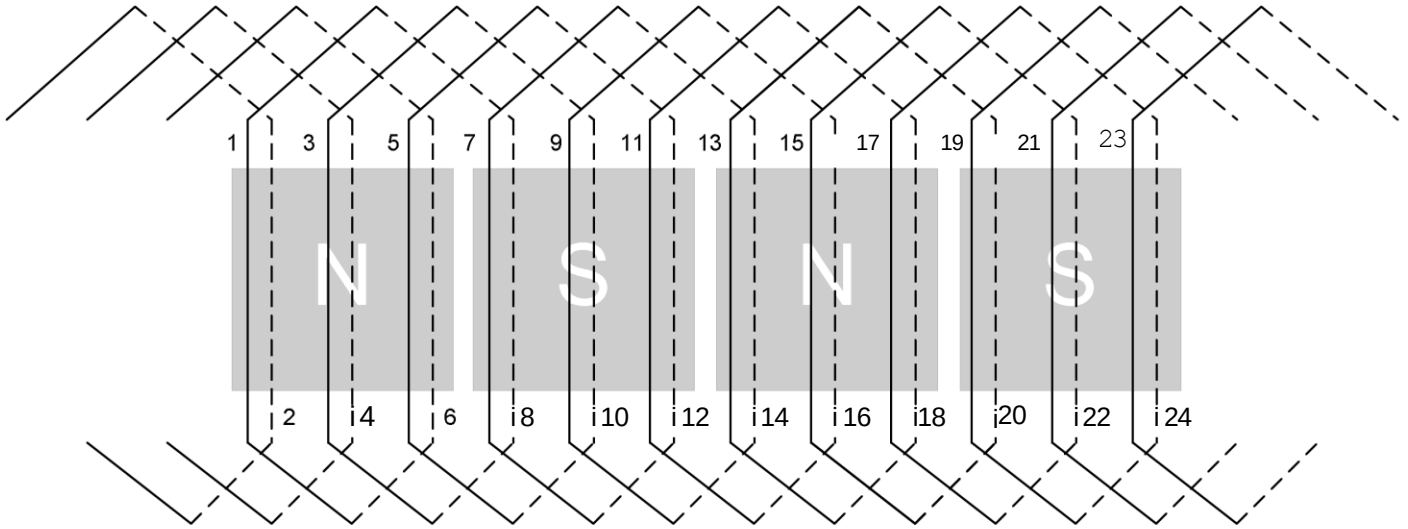
P = 4

Z = 12 x 2 = 24

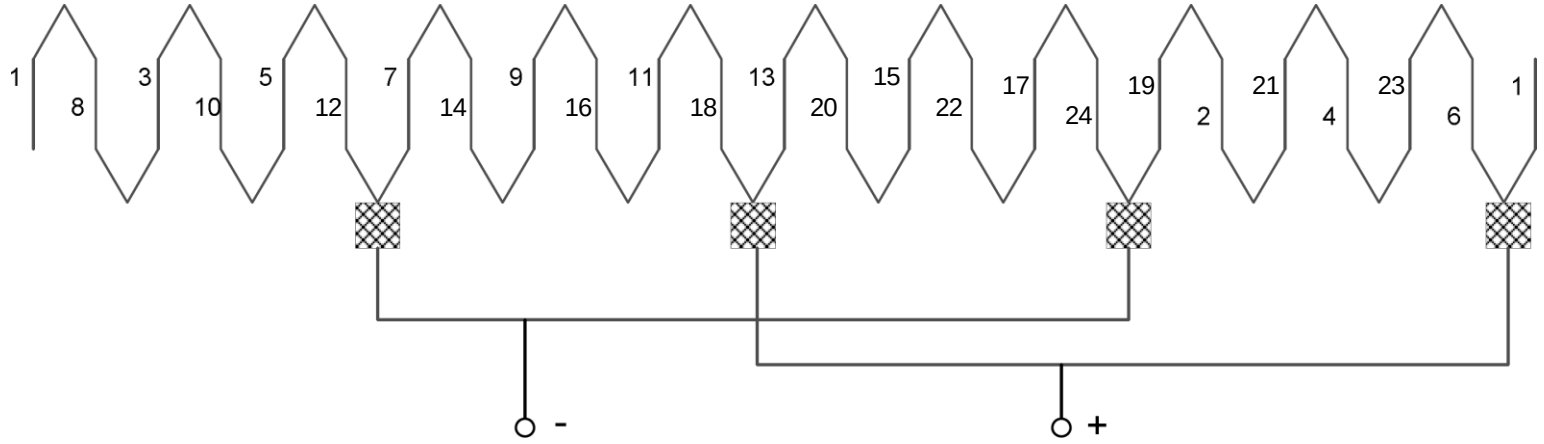
Select Back pitch $Y_B = 7$

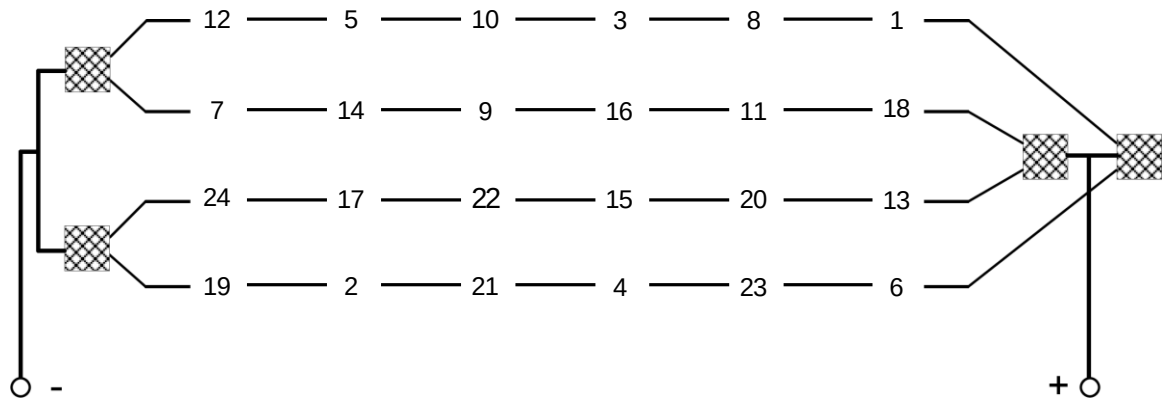
Front pitch $Y_F = 5$

Lap Winding Example



Lap Winding Example





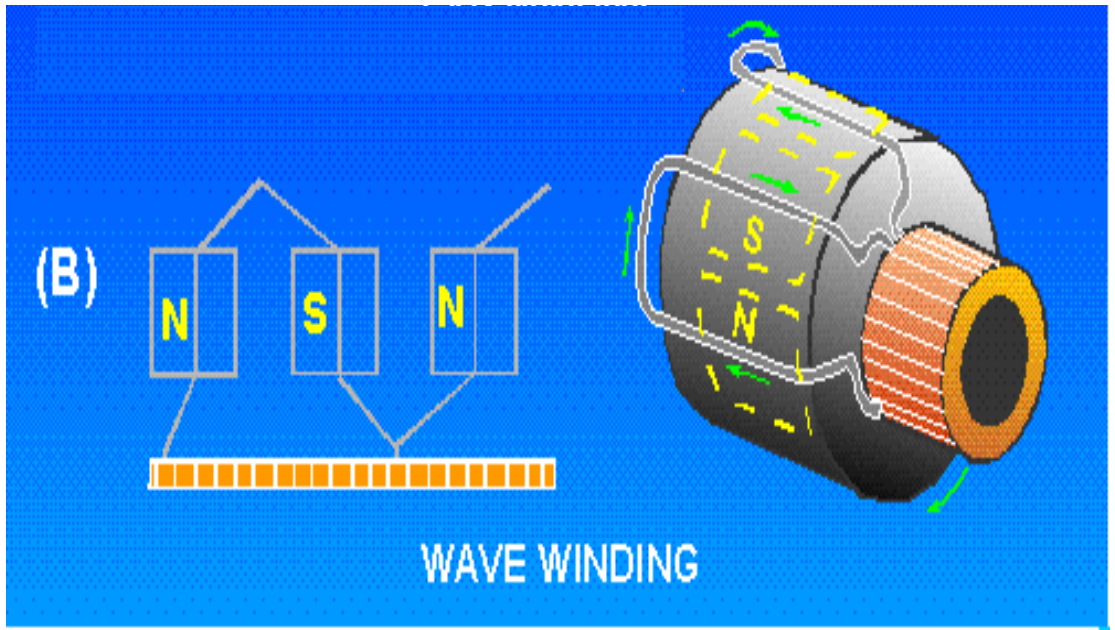
For simplex Lap winding,

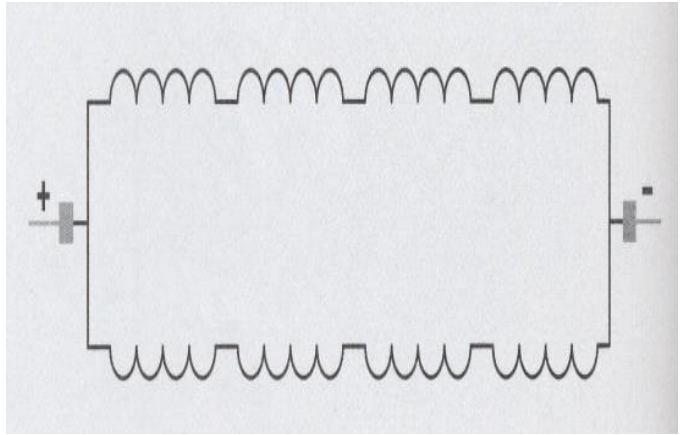
Number of parallel paths = P

P = *Number of poles*

ARMATURE WINDINGS (Cont)

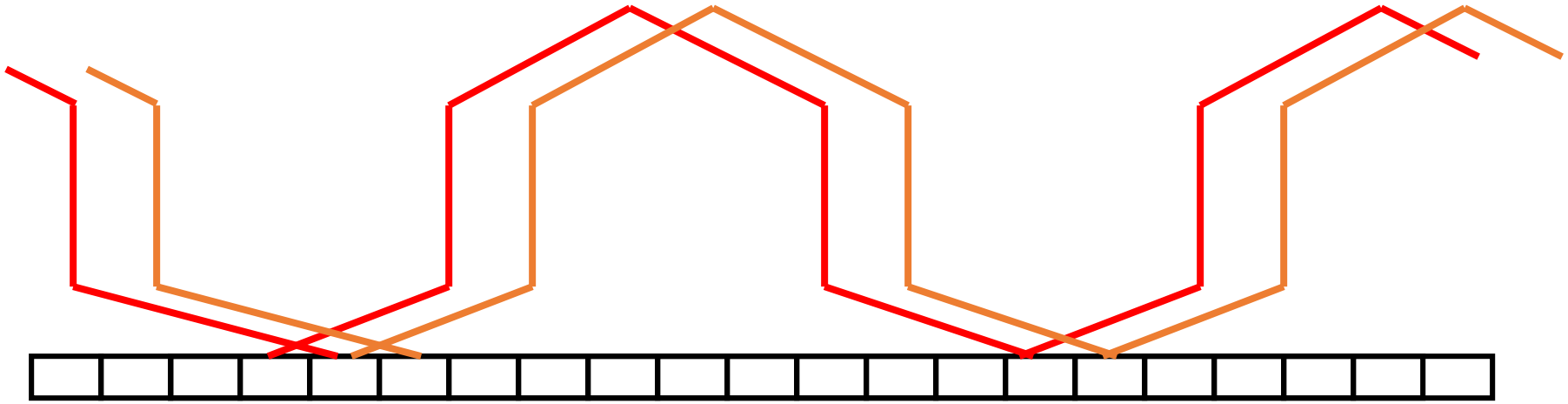
- Wave Wound Armatures
  - are used in machines designed for high voltage and low current
  - their windings connected in series
  - When the windings are connected in series, the voltage of each winding adds, but the current capacity remains the same
  - are used is in the small generator in hand-cranked megohmmeters
  - No of current path, C=2





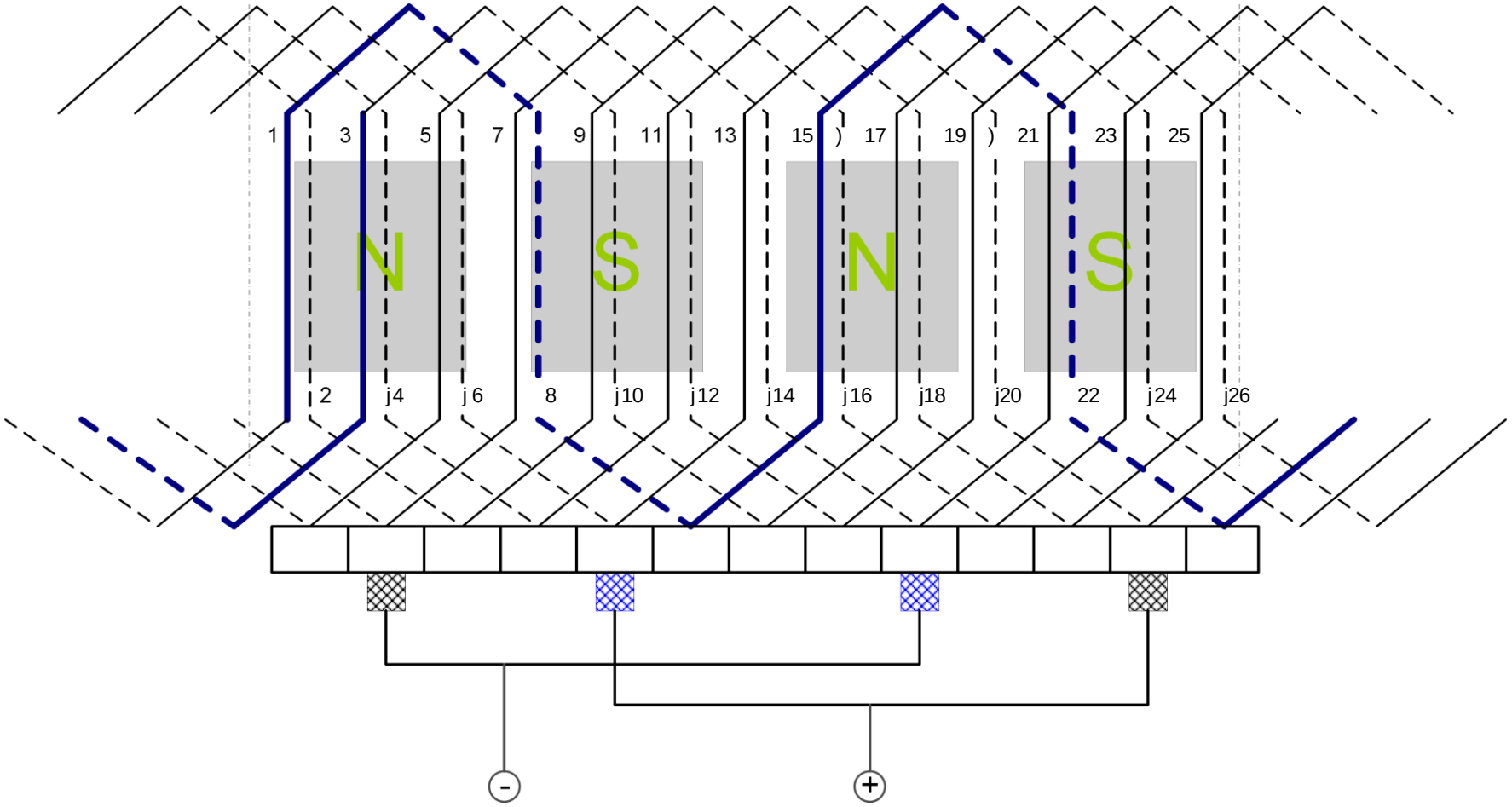
Elements of an armature windings

Wave winding



- p/2 coils in series between adjacent commutator bars
- ½ of all coils between brushes
- Regardless of no. of poles, there are always 2 parallel path
- The distance between end coils (commutator pitch) is $2(C \pm 1)/p$ where C is the no. of commutator bars

Wave Winding



Comparison between lap and wave windings

Since lap wound machine has a greater number of parallel path and lesser number of conductors per parallel path, it is suitable for low voltage high current applications	Wave wound machines are suitable for low current higher voltage applications
Lap wound machines require equaliser rings	Wave wound machines do not require equaliser rings since conductors in each parallel path pass through all the poles
Number of brushes is equal to the number of poles	Only two brushes are necessary but additional brushes may be added upto the number of poles

EMF generated in DC machines

Assume:

ϕ —useful ñux per pole

Z —total number of armature conductors

N —speed of the generator in revolutions per minute (rpm)

P —number of poles in the machine

A —number of parallel paths in the armature

Eg—generated emf

Since N is in rpm, Revolutions per second $= \frac{N}{60}$

Time taken for one revolution $= \frac{60}{N}$ seconds



EMF generated in DC machines

- Average e.m.f generated/conductor = $\frac{d\Phi}{dt}$ volt
- Now, flux cut/conductor in one revolution $d\Phi = \Phi P$ wb

No. of revolutions/sec=N/ 60

$\therefore$ Time for one revolution , dt= 60 /N sec

According to Faraday's Law of electro magnetic induction

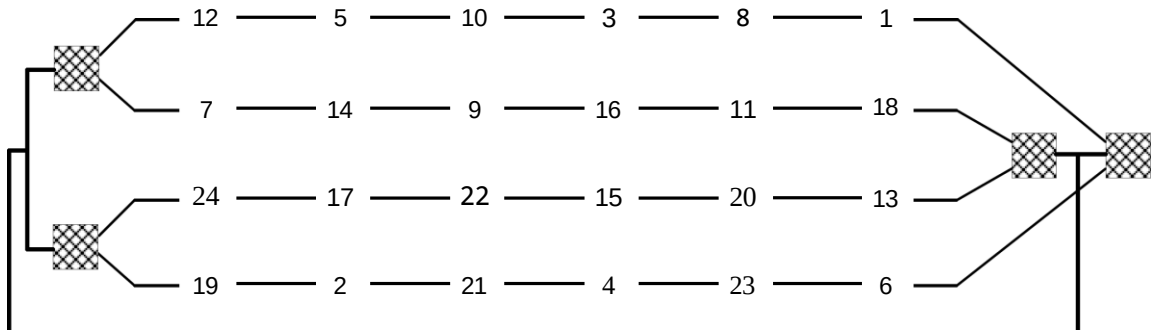
E.M.F generated/conductor = $\frac{d\Phi}{dt} = \frac{\Phi PN}{60}$ volts

No. of conductors (in series) in one parallel path= Z / A

$\therefore$ E.M.F generated/path= $\frac{\Phi PN}{60} \times \frac{Z}{A}$ Volts

$\therefore$ Generate E.M.F, $E_g = \frac{\Phi Z N}{60} \times \frac{P}{A}$ Volts

EMF generated in DC machines



Number of conductors in series in each path $= \frac{Z}{A}$

$$E = \frac{\Phi ZNP}{60A}$$

For

i) Wave winding A = 2
ii) Lap winding A = P

Example 1

A four-pole dc generator has 384 lap connected conductors. If the flux per pole is 25 mWb, find the induced voltage when the machine is running at 1500 rpm.

Since the machine is lap wound, $A = P = 4$

$$\text{Induced voltage, } E = \frac{\phi ZNP}{60A} = \frac{25 \times 10^{-3} \times 384 \times 1500 \times 4}{60 \times 4}$$

$$= 240 \text{ V}$$

Example 2

A four-pole wave wound dc machine has 256 armature conductors and the flux per pole is 0.02 Wb. At what speed the generator should run to generate a voltage of 220 V at its terminals without loading the machine?

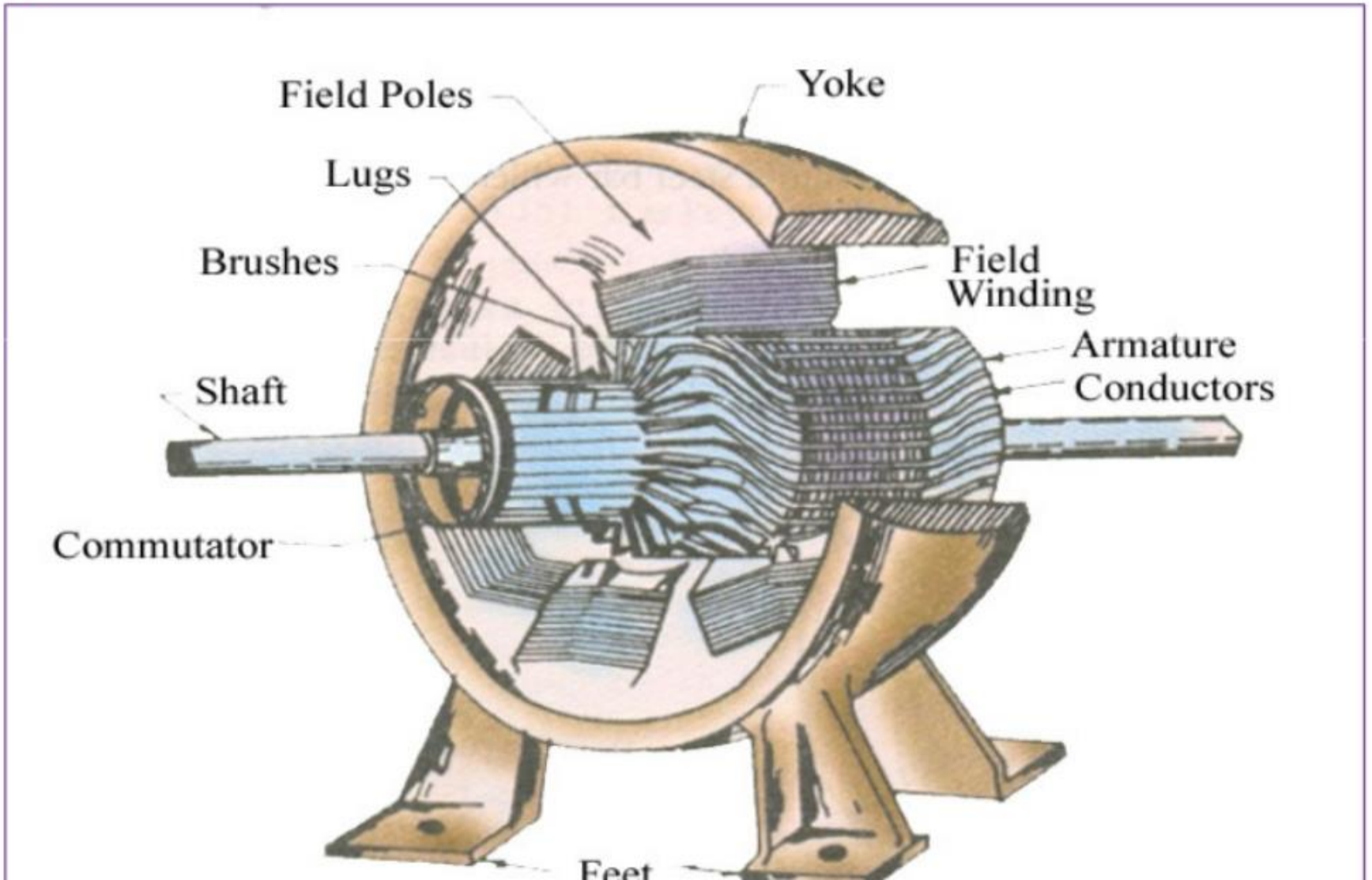
Since the machine is wave wound, $A = 2$

Generated voltage, $E = \dfrac{\phi ZNP}{60A}$

Therefore, speed, $N = \dfrac{60EA}{\phi ZP}$ $\dfrac{220 \times 60 \times 2}{0.02 \times 256 \times 4} = 1289$ rpm

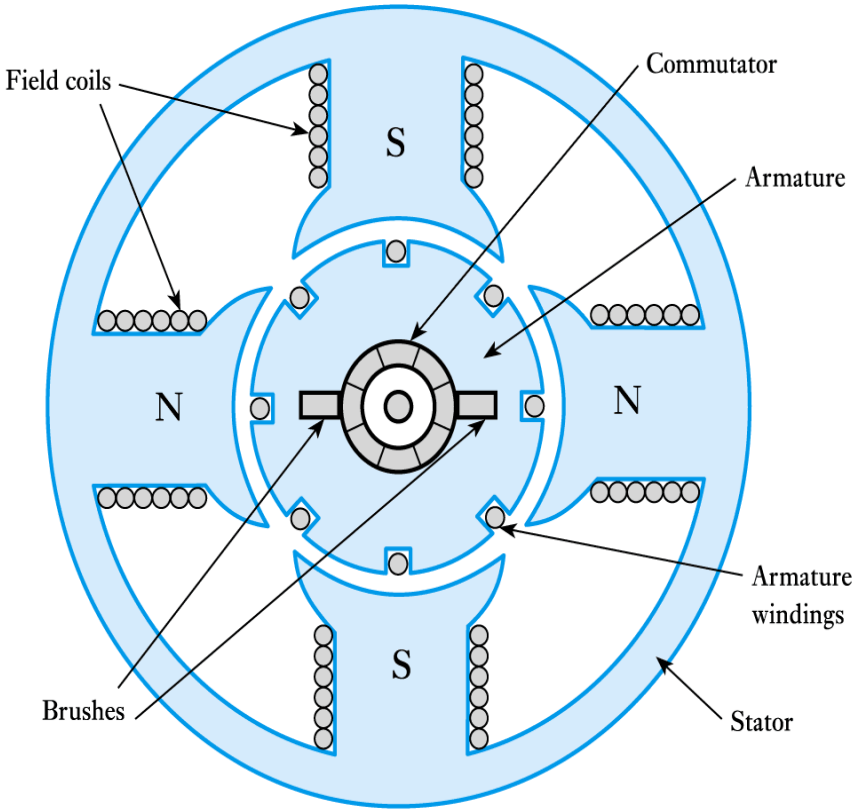
Constructional details of DC machines

Main construction diagram



Construction of DC Generator

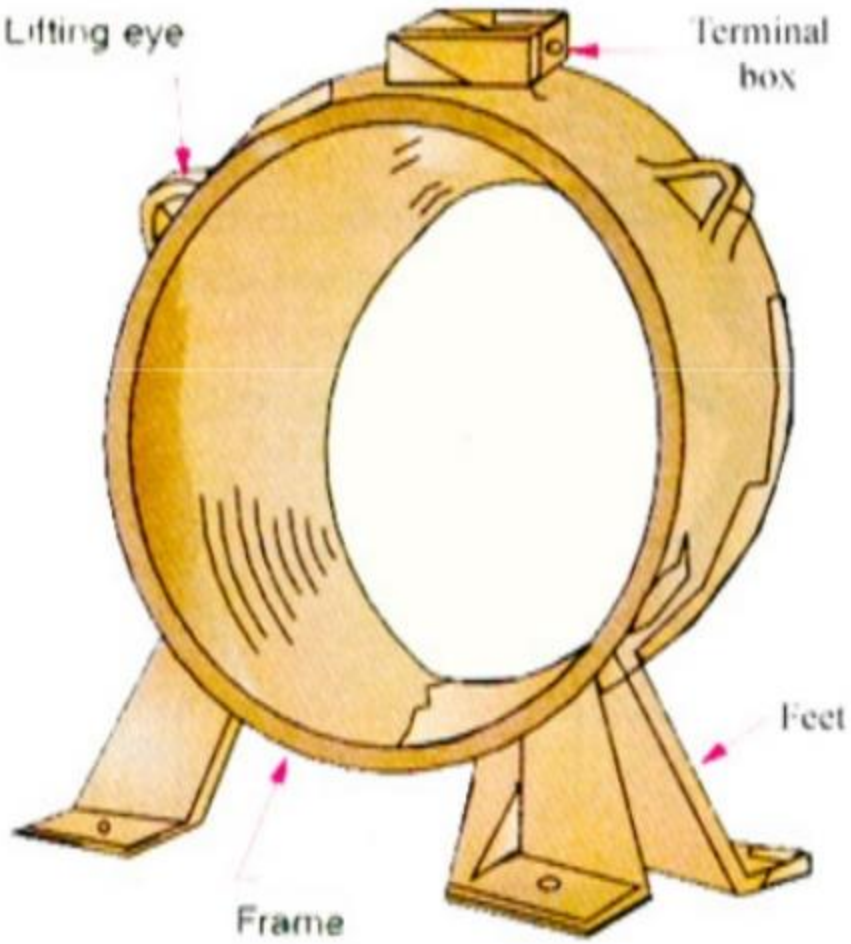
- Yoke
- Field core and coils
- Armature core
- Armature winding
- Commutator
- Brushes



Yoke

- Its main function is to form a protective covering over the inner sophisticated parts of the motor and carries the flux produced by the poles.

- Made up of cast iron or steel.



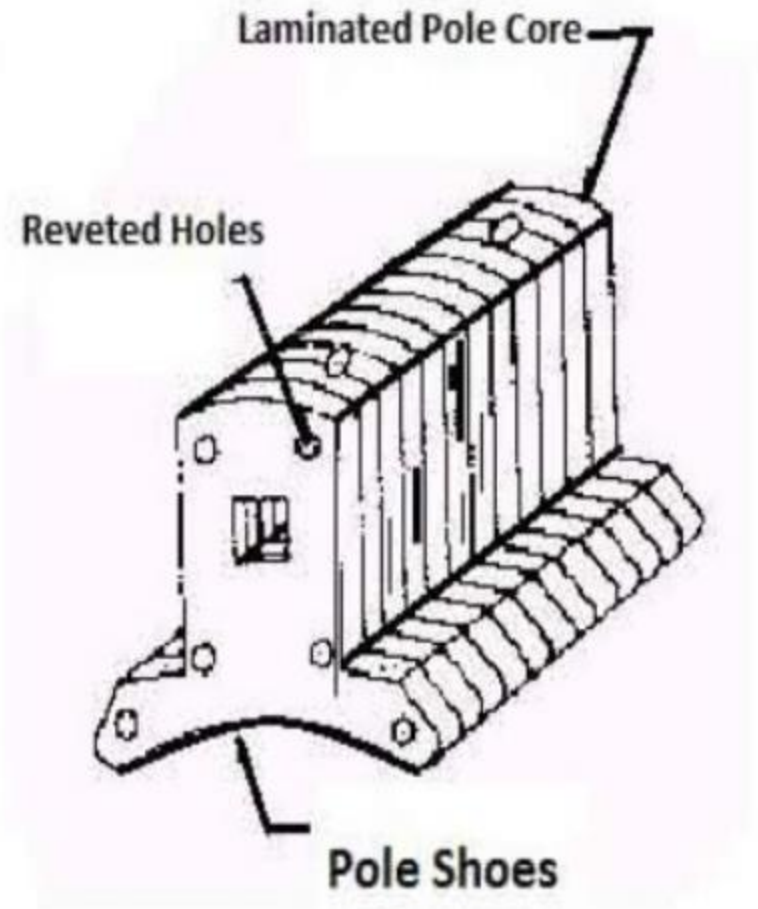
Field system

- It is for uniform magnetic field within which the armature rotates.
- Electromagnets are preferred in comparison with permanent magnets
- They are cheap , smaller in size , produce greater magnetic effect and
- Field strength can be varied

Poles

- The magnetic poles are structures fitted onto the inner wall of the yoke with screws.

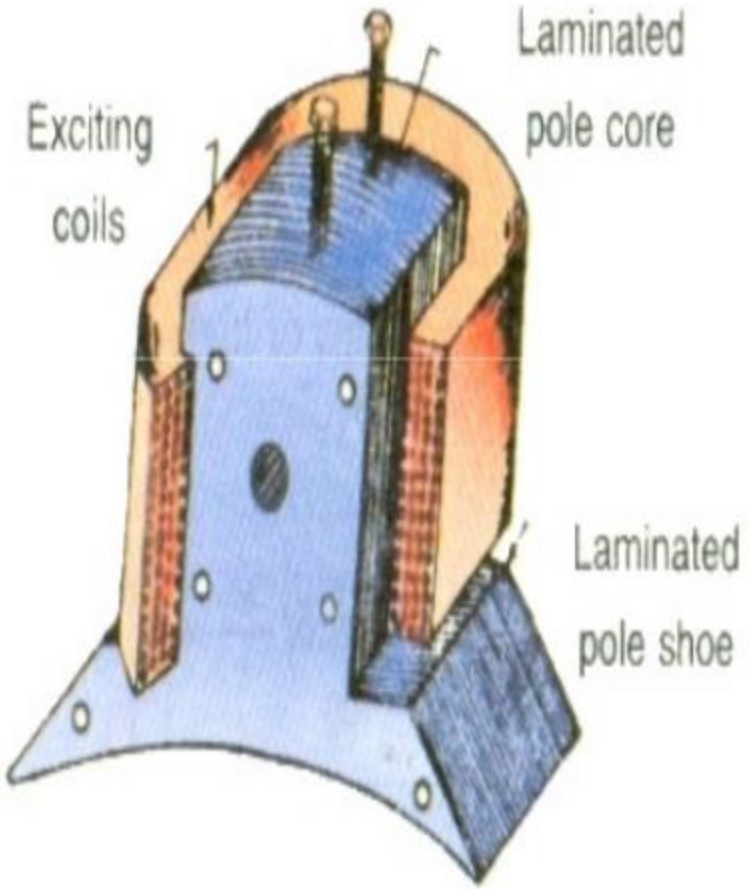
- The construction of magnetic poles basically comprises of two parts namely, the pole core and the pole shoe stacked together under hydraulic pressure and then attached to the yoke.



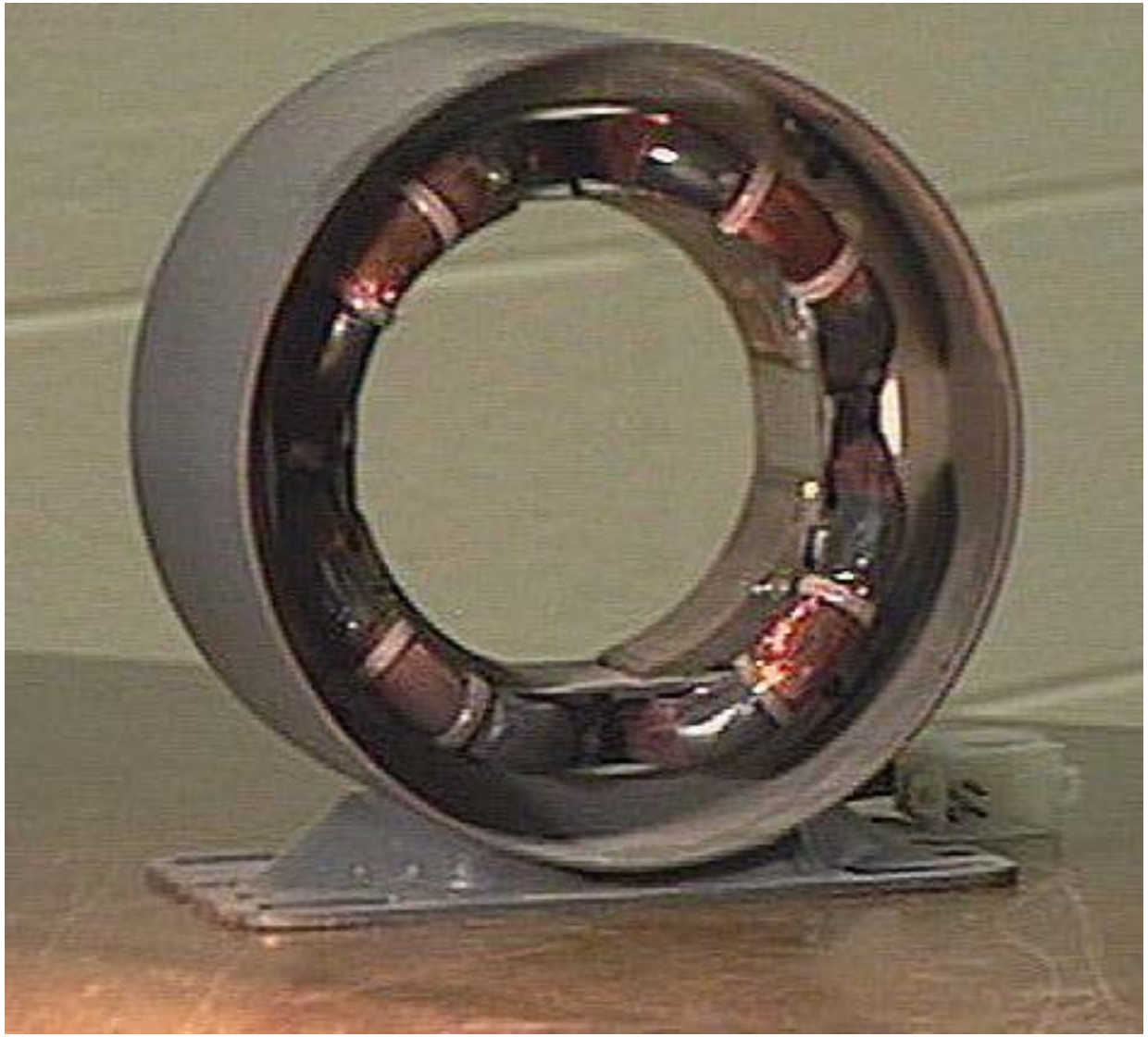
Field winding

- Made with field coils (copper wire) wound over the slots of the pole shoes

- The field windings basically form an electromagnet, that produces field flux



Field winding

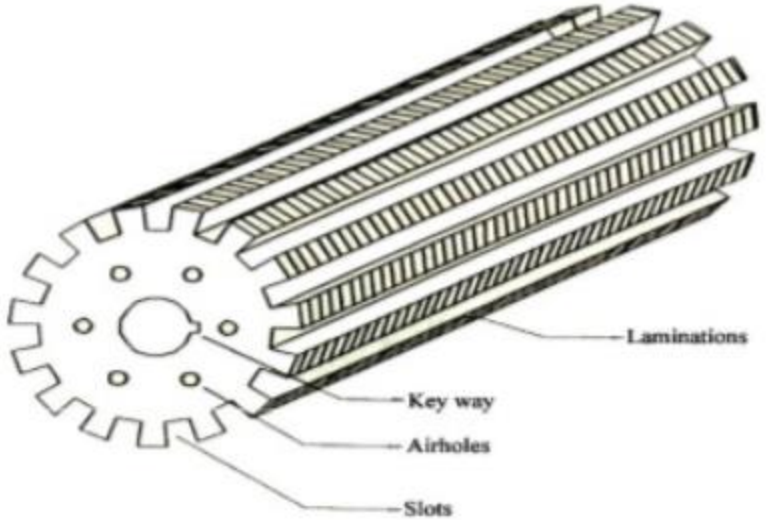


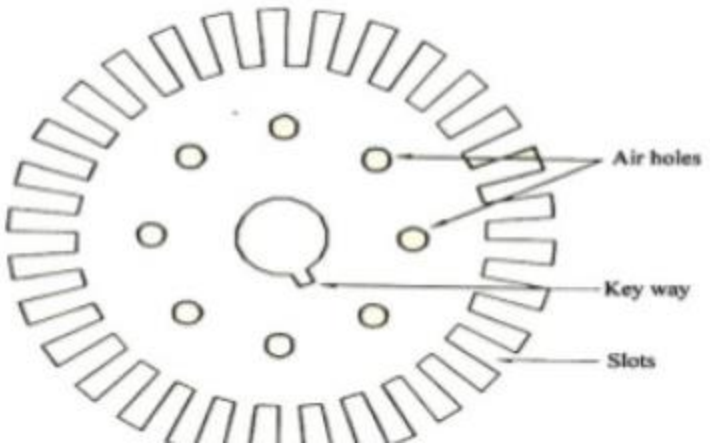
Armature core (Rotor)

- The armature core is cylindrical
- High permeability silicon steel stampings
- Impregnated
- Lamination is to reduce the eddy current loss

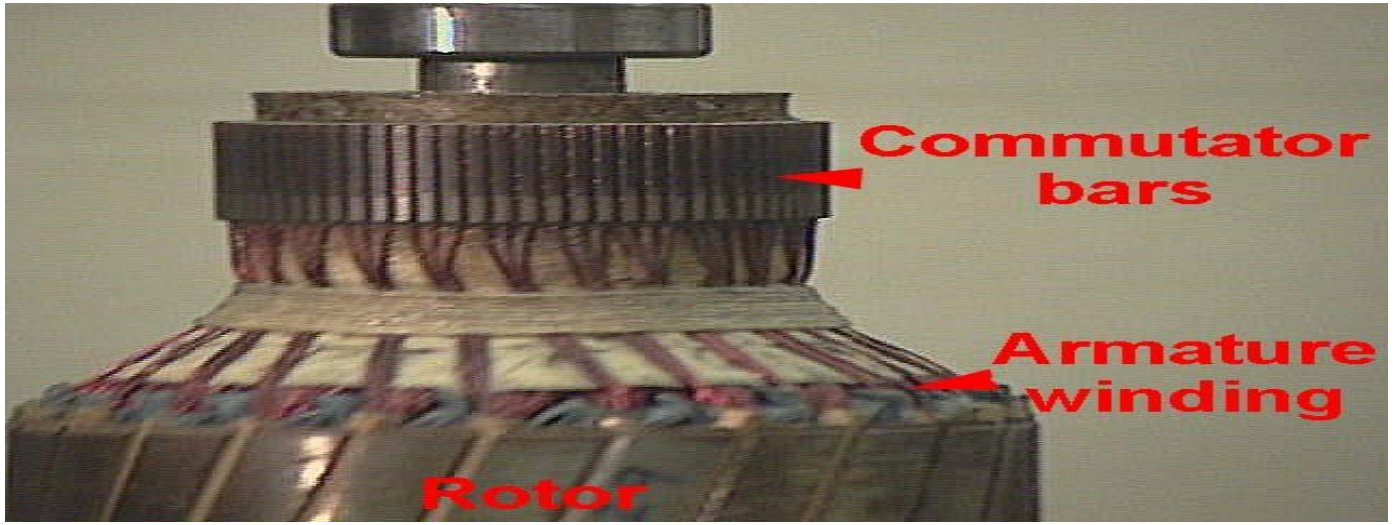
Armature

- The armature core are provided with slots to which the armature windings made with several turns of copper wire distributed uniformly over the entire periphery of the core.





Rotor and rotor winding





Commutator

- Connect with external circuit
- Converts ac into unidirectional current
- Cylindrical in shape
- Made of wedge shaped copper segments
- Segments are insulated from each other
- Each commutator segment is connected to armature conductors by means of a cu strip called riser.
- No of segments equal to no of coils

Commutator

- It is a cylindrical structure made up of copper segments stacked together, but insulated from each other by mica.
- Collects the current from the armature conductors
- Converts alternating current induced in the armature circuit into unidirectional current.



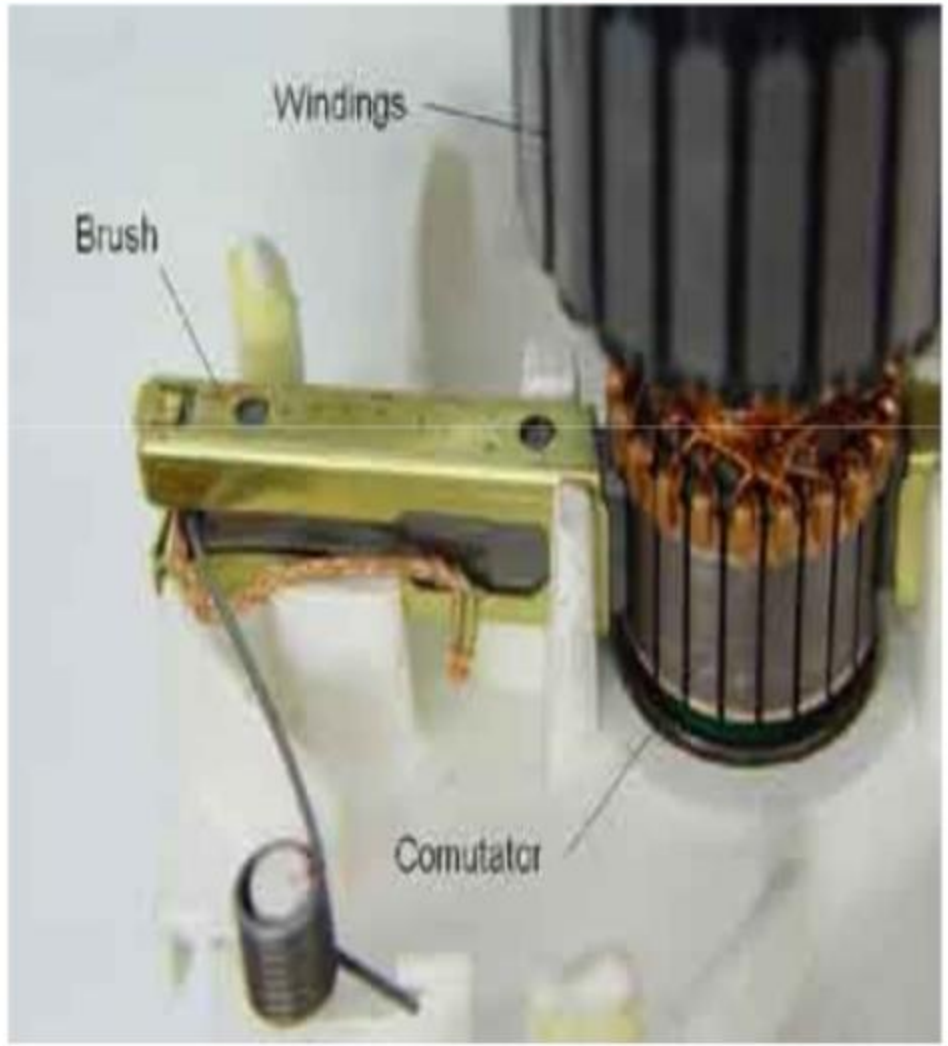
Carbon brush

- Carbon brushes are used in DC machines because they are soft materials
- It does not generate spikes when they contact commutator
- To deliver the current thro armature
- Carbon is used for brushes because it has negative temperature coefficient of resistance
- Self lubricating , takes its shape , improving area of contact

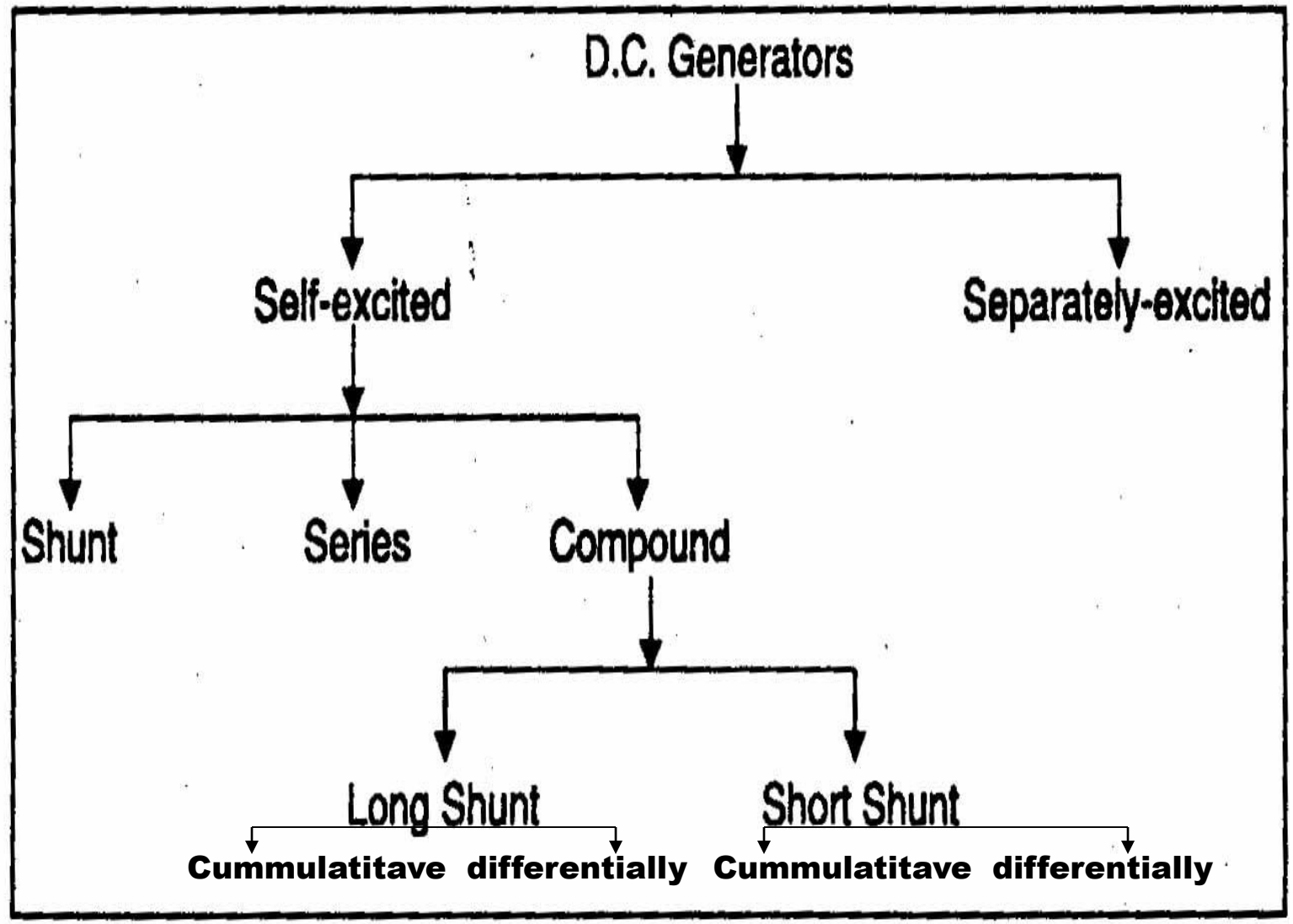
Brushes

- Made with carbon or graphite structures, making sliding contact over the rotating commutator.

- Collects the current from commutator and make it available to the stationary ckt.



Classification of DC generator


D.C. Generators
Self-excited
Separately-excited
Shunt
Series
Compound
Long Shunt
Short Shunt
Cummulatitave differentially
Cummulatitave differentially

Types of Generators

1)Separately excited generators

2)Self excited generators

i) shunt wound

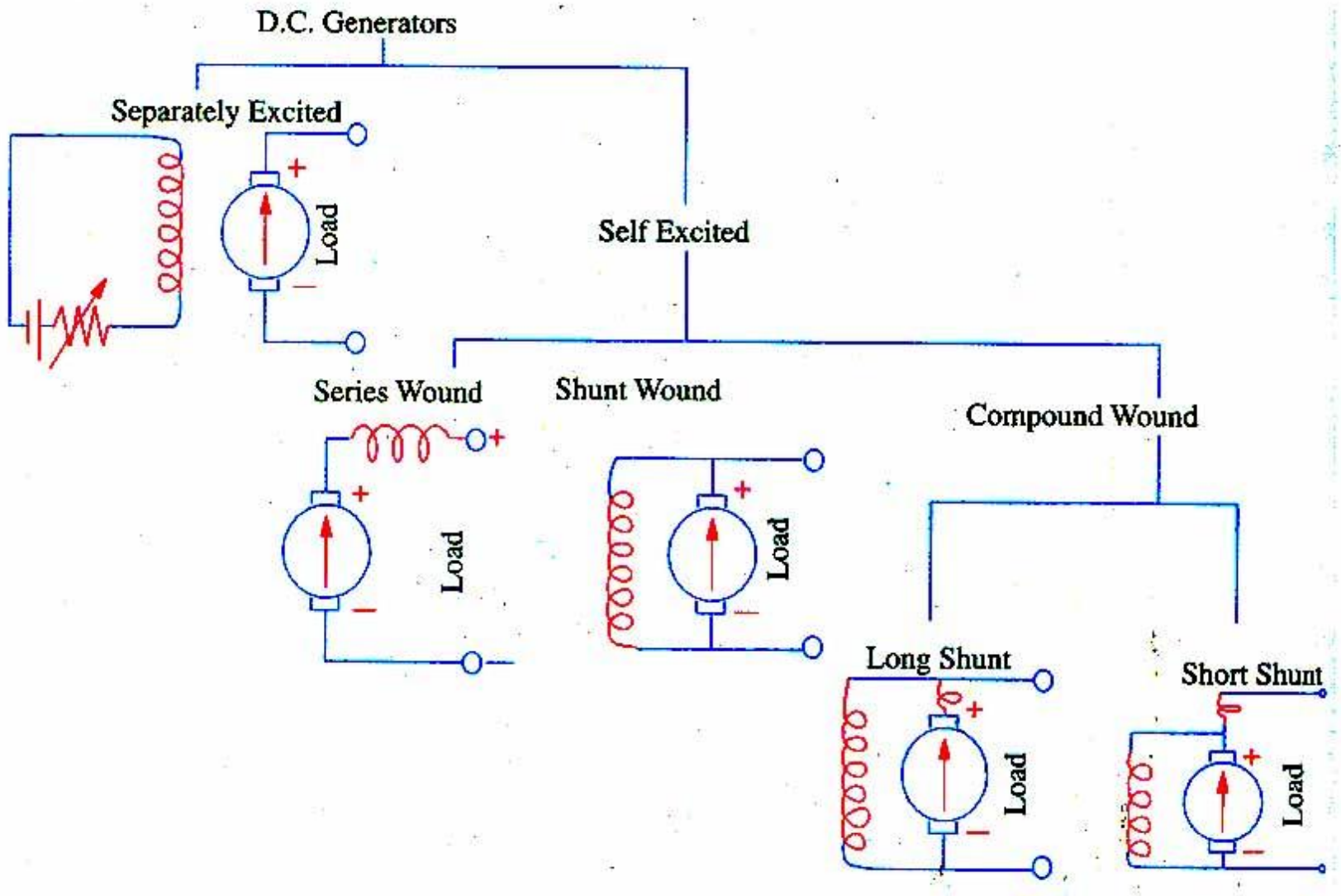
ii) series wound

iii) compound wound

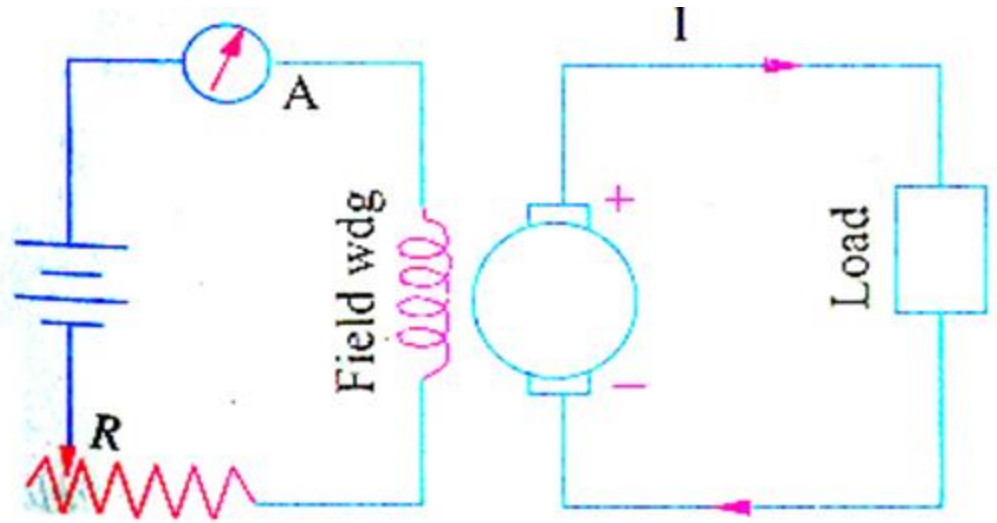
a) long shunt

b) short shunt

Clasifications of Generators



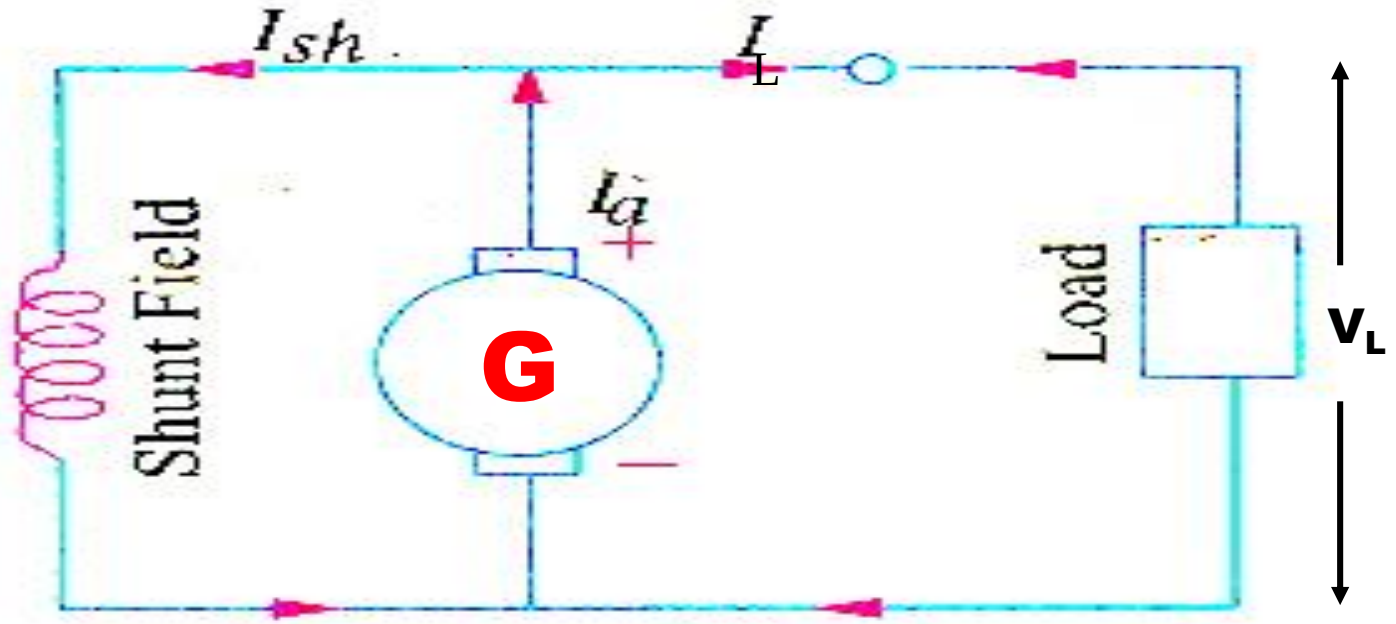
Separately excited generators



$I_a = I_L$

$E = V_t + I_a R_a + BCD$

shunt wound



Shunt generator

(a)

$I_L = I_a - I_{sh}$

$I_a = I_L + I_{sh}$

$\in g = V_L + I_a R_a$ + B.C.D (Brush contact drop)

$V_L = R_{sh} \cdot I_{sh} = I_L \cdot R_L$

$I_L = \frac{\text{Load}}{V_L}$ (in watts)

R_L = load resistan

series wound



Series generator

(b)

$$I_L = I_a = I_{se}$$

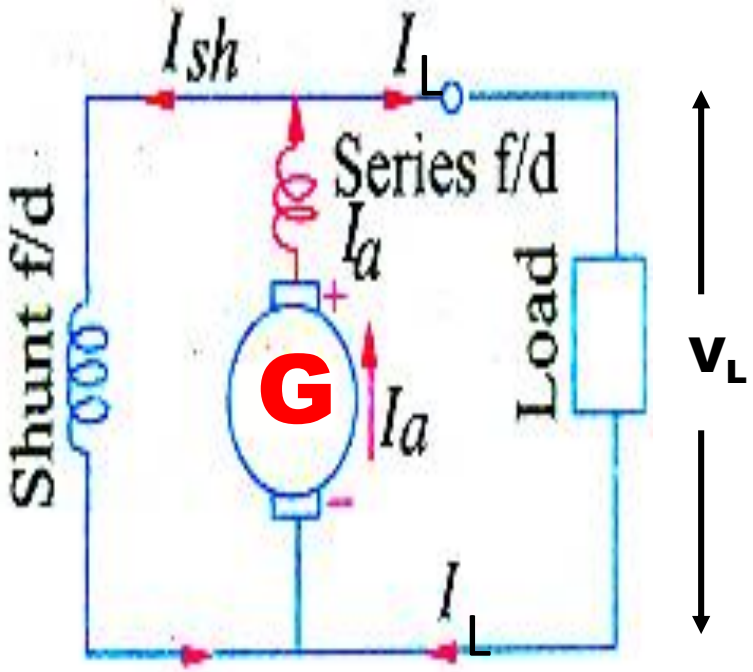
$$Eg = V_L + I_a R_a + I_{se} R_{se}$$

$$= V_L + I_a (R_a + R_{se})$$

$$V_L = I_L \cdot R_L$$

compound wound

long shunt



© Long shunt

$I_a = I_{se} = I_L + I_{sh}$

$I_{sh} = \frac{V_L}{R_{sh}}$

$E_g = V_L + I_a R_a + I_{se} R_{se}$

$= V_L + I_a (R_a + R_{se})$

$I_{sh} = \frac{E_g - I_a (R_a + R_{se})}{R_{sh}}$

compound wound

short shunt

(d) SLOW shunt.

$I_L = \frac{P_L}{V_L}$ (load in watts) , $I_L = I_{se}$

$I_{sh} = \frac{V_L + I_{se} R_{se}}{R_{sh}}$, $I_a = I_L + I_{sh}$

$E_g = V_L + I_a R_a + I_{se} R_s$

